Jongwoon Lee

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RESEARCH INTERESTS

Domain shift and out-of-distribution generalization · continual learning · weakly supervised learning for medical imaging · adaptation of vision foundation models · structured prediction with language models. I build machine learning methods that hold up when the deployment hospital, scanner, or instrument is not the one the model was trained on.

I started in computational pathology during my PhD in Medicine, then entered the Department of Brain and Cognitive Engineering at Korea University to begin a full-time engineering career. Seven years of clinical pathology practice give me access to multi-hospital data and the judgment to evaluate what actually transfers.

EDUCATION

M.S., Computer Science, University of Rochester Jan 2026 – May 2027 (expected)

- Cumulative GPA 3.91 / 4.00.

Ph.D. Candidate (on leave of absence), Brain and Cognitive Engineering, Korea University Sept 2024 –

- Advisor: Prof. Dong-Joo Kim. 25 credits completed — Pattern Recognition, Computer Vision, Machine Learning, Mathematical Foundations of AI; GPA 4.5 / 4.5.

Ph.D., Medicine, University of Ulsan College of Medicine, Seoul, KR 2022 – 2024

- Advisor: Prof. Hee Jin Lee, Dept. of Pathology, Asan Medical Center.

M.Sc., Medicine, University of Ulsan College of Medicine, Seoul, KR 2019 – 2021

- Advisor: Prof. Jihun Kim, Dept. of Pathology, Asan Medical Center. Admitted first in class.

M.D., University of Ulsan College of Medicine, Seoul, KR 2010 – 2017

The M.D., M.Sc. and Ph.D. above are three separately conferred Korean degrees, not a combined M.D./Ph.D. program.

SELECTED RESEARCH EXPERIENCE

Graduate Research Assistant — Lab of Prof. Sean Crowell, University of Rochester Sept 2026 –

- Cross-instrument transfer and domain shift in remote-sensing CO_2 retrieval.

Structured Prediction with Language Models for Electrodiagnostic Interpretation May 2026 –

University of Rochester — Dept. of Neurology and Lab of Prof. Jiebo Luo, Dept. of Computer Science

- Built a two-stage pipeline that turns free-text EMG/NCS studies into structured diagnostic slots and then back into a natural-language report, composing six domain-specialist models through a rule-based arbiter.
- Reached diagnostic-level F1 of 0.89 on 404 held-out studies, against 0.84 for a single fine-tuned baseline; ablations isolate which composition step produces each gain.
- First author; submitted to *AAAI-27*. Clinical collaborators: Drs. E. Logigian and J. Nutt, University of Rochester Medical Center.

Cross-Hospital Adaptation of Pathology Foundation Models 2025 –

University of Rochester / Korea University Guro Hospital

- Measured which interventions actually transfer frozen UNI and Virchow2 features across three hospital cohorts (Asan, Guro, BCI) for Nectin-4, HER2 and TROP2 immunohistochemistry grading.
- Developed Drift-Aware Spectral Gating (DASG), a principal-component–gated low-rank input transform that raises target-hospital macro-F1 by 0.125 over the RGB baseline without any target labels.
- Added a mechanistic analysis of how the gating overlaps with feature mean-shift correction.
- First author; accepted at *MICCAI CLINICCAI 2026* (to be presented September 2026); the full method, with expanded slide-level and prevalence-shift evaluation, is in preparation for *WACV 2027*.

Weakly Supervised Prognostic Modeling in Digital Pathology 2022 –

Asan Medical Center / Korea University Guro Hospital

- Attention-based multiple-instance learning on whole-slide images for recurrence and survival prediction in bladder, colorectal, oral tongue and bile duct carcinoma.
- Tumor-infiltrating-lymphocyte quantification validated against clinical outcome across four tumor types.
- Principal Investigator on a ₩10,000,000 institutional grant; this line of work produced five first- or corresponding-author papers and a Best Poster Prize at the European Congress of Pathology 2025.

SOFTWARE

fft-lowfreq-adaptation — Reference implementation of FFT-based frequency adaptation for cross-hospital IHC expression grading with frozen pathology foundation models (CLINICCAI at MICCAI 2026). `github.com/jongwoonalee/fft-lowfreq-adaptation`

PEER-REVIEWED PUBLICATIONS

** first author † corresponding author*

- Jo D, **Lee J***, et al. Field-of-View Impact on Multi-Class Tissue Segmentation: A Systematic Analysis of Size-Dependent Detection Performance Across Multiple Cancer Types. *Laboratory Investigation*, 2026.
- Lee S, **Lee J**†, et al. Artificial Intelligence-Assessed Tumor Central Tumor-Infiltrating Lymphocytes Predict Recurrence-Free Survival in Transurethral Resection of Bladder Tumor Specimens. *Laboratory Investigation*, 2026.
- **Lee J***, et al. Assessment of Established Prognostic Factors and Artificial Intelligence-based Evaluation of Tumor-infiltrating Lymphocytes in Oral Tongue Squamous Cell Carcinoma. *Oral Oncology*, 2025. Discussed in an invited commentary: *Oral Oncology* 2025;168:107586.
- **Lee J***, et al. Factors Associated with Engraftment Success of Patient-Derived Xenografts of Breast Cancer. *Breast Cancer Research*, 2024.
- **Lee J***, et al. High-Grade Transformation of a Pancreatic Neuroendocrine Microtumor in a Patient with Von Hippel-Lindau Syndrome: A Case Report. *Journal of Pathology and Translational Medicine*, 2024.
- **Lee J***, et al. Extremely Well-Differentiated Adenocarcinoma of the Stomach: Diagnostic Pitfalls in Endoscopic Biopsy. *Journal of Pathology and Translational Medicine*, 2022.
- **Lee J***, et al. Breast Implant-Associated Anaplastic Large Cell Lymphoma: The First South Korean Case. *Journal of Pathology and Translational Medicine*, 2020.
- **Lee J***, et al. Enlarged Parent Artery Lumen at Aneurysmal-Neck Segment in Wide-Necked Distal Internal Carotid Artery Aneurysms. *Neurointervention*, 2015.

MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

- **Lee J***. Where to Decompose: Expert Composition and Structured Prediction for Electrodiagnostic Interpretation. Under review, *AAAI-27*.
- **Lee J***. Drift-Aware Spectral Gating for Cross-Hospital Transfer of Frozen Pathology Foundation Models. In preparation for *WACV 2027*.
- Yang Y, Jung D, **Lee J***, et al. The Expression and Clinical Relevance of LRRC15+ Cancer-Associated Fibroblasts in Colorectal Cancer. Under review, 2026.
- **Lee J***, et al. Extranodal Extension as a Prognostic Factor in Distal Bile Duct Carcinoma: Artificial Intelligence-Based Evaluation of E-cadherin Loss Associated with Tumor Invasiveness. Under review, 2026.
- Kim HS, **Lee J**†, et al. End-to-End Artificial Intelligence Model Identifying High Nectin-4 Expression from H&E Images as Low-Recurrence Risk in Non-Muscle Invasive Bladder Cancer. Under review, 2026.

MENTORING

Mentor — residents and medical students in computational pathology, Korea University Guro Hospital 2024 – 2025

CLINICAL AND PROFESSIONAL EXPERIENCE

Clinical Assistant Professor, Equivalent, Dept. of Pathology, Korea University Guro Hospital, KR Mar 2024 – Aug 2025

- Led deep learning research and contributed to scientific publications on computational pathology.
- Served as an external evaluator of breast cancer metastatic lymph node segmentation systems for DeepBio and AIVIS Co.

Resident and Fellow, Dept. of Pathology, Asan Medical Center, KR Mar 2018 – Feb 2024

- Chief Resident for six months, then Chief Fellow for a further six months.
- Organized and led 70+ interdepartmental and intradepartmental conferences.

Board-certified in Pathology (Korea, 2022); 7 years of clinical genitourinary pathology practice.

INVITED TALKS

- Genitourinary Pathology in 2025: Key Updates and Future Directions. Korean Society of Pathologists Spring Conference, Seoul, 2025.
- Ancillary Tests for Urinary Cytology Diagnosis in the Era of the Paris System. Korean Society of Cytopathology Spring Conference, Seoul, 2025.

- Panel discussant, "Sharing Experiences with TPS and Strategies for Promotion," Korean Society of Cytopathology Spring Conference, Seoul, 2025.
- Case discussant, subcommittee session GU-J2, Korea–Japan International Academy of Pathology, Kitakyushu, Japan, 2025.

SELECTED CONFERENCE PRESENTATIONS

- Separating Scanner from Signal: FFT-Based Low-Frequency Adaptation for Cross-Hospital IHC Expression Grading. *CLINICCAI at MICCAI*, Strasbourg, France, September 2026 (upcoming). (Poster, first author)
- Field-of-View Impact on Multi-Class Tissue Segmentation. *USCAP*, Texas, USA, 2026. (Platform, first author)
- End-to-End AI Model Identifying High Nectin-4 Expression from H&E Images in Non-Muscle Invasive Bladder Cancer. *USCAP*, Texas, USA, 2026. (Poster, corresponding author)
- AI-Assessed Tumor Central Tumor-Infiltrating Lymphocytes Predict Recurrence-Free Survival in TURBT Specimens. *USCAP*, Texas, USA, 2026. (Poster, corresponding author)
- AI Prediction of Clinicopathological Factors and LRRC15-Immunolabeled Cancer-Associated Fibroblast Level in Colorectal Carcinoma Using Attention-Based Multiple Instance Learning. *37th European Congress of Pathology*, Vienna, Austria, 2025. **First Best Poster Prize.**
- Weakly Supervised Learning for Predicting Recurrence and Invasion in Bladder Cancer from Transurethral Resection Specimens. *USCAP*, Boston, USA, 2025. (Poster)
- Evaluation of Prognostic Factors for Oral Tongue Squamous Cell Carcinoma. *ASCO Breakthrough*, Yokohama, Japan, 2024. (Poster)
- AI-Assisted Tumor-Infiltrating Lymphocytes as Predictors of Tumor Recurrence in TURBT. *Asian Society of Digital Pathology*, Seoul, KR, 2024. (Poster)
- Epithelial-to-Mesenchymal Transition in the Extranodal Extension of Distal Extrahepatic Bile Duct Carcinoma. *76th Korean Society of Pathologists Fall Meeting*, Seoul, KR, 2024. (International poster)
- The Principal Factors Associated with PDX Engraftment in Breast Cancer. *San Antonio Breast Cancer Symposium*, Texas, USA, 2022 and 2023. (Posters)
- Ten additional posters and platform presentations at Korean Society of Pathologists / Korean Society of Cytopathology meetings, 2018–2023; full list available on request.

RESEARCH FUNDING

Principal Investigator — "Weakly Supervised Learning for Prognostic Modeling in Bladder Cancer using Multiple Instance Learning," 2025 Research Initiative Grant, Korea University Guro Hospital (₩10,000,000, approx. USD 7,000) 2025

HONORS AND AWARDS

First Best Poster Prize, 37th European Congress of Pathology (€1,000)	2025
USCAP International Conference Travel Grant, Korean Society of Pathologists	2024
Excellence in Research Award, Asan Medical Center	2021
Best Abstract Award, Korean Society of Cytopathology Fall Meeting	2020
Admitted first in class, M.Sc. in Medicine, University of Ulsan College of Medicine	2019
Medical School Scholarship Type A, University of Ulsan Graduate School of Medicine (8 semesters)	2019 – 2023
Merit-based scholarships, University of Ulsan College of Medicine (7 semesters)	2012 – 2015

PROFESSIONAL SERVICE

Reviewer, *COMPAYL* (Computational Pathology and Multimodal Data Workshop), MICCAI	2026
Reviewer, *The Breast Journal*	2025, 2026
Reviewer, *International Journal of Breast Cancer*	2026
Reviewer, *Journal of Pathology and Translational Medicine*	2025
Abstract Reviewer, Korean Society of Pathologists International Congress	2025
Information Committee Member, Korean Society of Pathologists	2024 –
Item writer, Korean Pathology Board Examination	2025
Case contributor, Genitourinary Pathology E-learning Center for pathology residents	2025

TECHNICAL SKILLS

Languages: Python (primary), R

Frameworks: PyTorch, HuggingFace Transformers, tensorFlow

Methods: Multiple-instance learning, attention-based aggregation, domain adaptation, survival modeling, LLM fine-tuning (LoRA / supervised fine-tuning), structured prediction

Tools and Infrastructure: QuPath, OpenSlide, multi-GPU training

CERTIFICATIONS AND MEMBERSHIPS

Board Certification in Pathology, South Korea (2022) • Medical License, South Korea (2017)

Member: United States and Canadian Academy of Pathology; Korean Society of Pathologists; Genitourinary, Molecular and Breast Pathology Study Groups.